SAMYA SAMANTA

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EDUCATION

Ph.D. in Chemistry, Texas A&M University, College Station, TX 2021 – 2026
Dissertation (defended): Electrophilic Amine Salts in Nitrogen-Group Transfer Reactions
Advisor: David C. Powers | GPA: 4.00 / 4.00

BS & MS in Chemistry, Indian Institute of Technology Kharagpur, India 2015 – 2020
GPA: 9.02 / 10.00

PROFESSIONAL SUMMARY

Synthetic organic chemistry researcher with a defended Ph.D. dissertation and more than three years of laboratory experience. Specializes in reaction development, cross-electrophile coupling, photoredox chemistry, and mechanistic investigations. Develops reproducible synthetic protocols, evaluates reaction conditions and substrate scope, and communicates results through publications and presentations. Experienced in retrosynthetic analysis, purification, analytical characterization, and mentoring researchers.

TECHNICAL SKILLS

Synthetic Chemistry & Reaction Development: Organic synthesis • Retrosynthetic analysis • Synthetic route design • Reaction optimization • Method development • Cross-electrophile coupling • Photoredox chemistry • Transition-metal catalysis • Electrochemistry • Radical chemistry • Mechanistic investigations • Air-sensitive synthesis

Analysis & Purification: HPLC • NMR • GC-MS • EPR • IR • UV-Vis • Cyclic voltammetry • Column chromatography • Purification • Compound characterization

Experimental Practice & Knowledge Sharing: Experimental design • Reaction-condition and substrate-scope evaluation • Experimental troubleshooting • Reproducible protocol development • Technical documentation • Scientific presentations and publications • Laboratory safety • Mentoring and training

RESEARCH EXPERIENCE

Texas A&M University, College Station, TX, USA *(Doctoral Researcher)* **2021 – 2026**

- Established a base-free C–N cross-electrophile coupling strategy using *N*-centered radicals, enabling orthogonal reactivity for C–N bond formation (*J. Am. Chem. Soc.* **2026**, 148, 33104–33117).
- Established a robust, scalable, and reproducible synthetic protocol for *N*-pyridinium aziridines, a bench-stable aziridine synthon; documented the procedure in *Org. Synth.* **2026**, 103, 78–100.
- Developed two complementary aminyl radical generation platforms based on redox-active *N*-pyridinium and *N*-phenothiazinium amine salts, expanding the toolbox for *N*-centered radical generation.
- Elucidated oxidative and reductive pathways for photogeneration of arylamine radicals from *N*-pyridinium arylamine precursors, informing mechanistic understanding of nitrogen-group transfer chemistry (*ACS Catal.* **2024**, 14, 13156–13162).
- Mentored four undergraduate and two graduate researchers in organic synthesis, purification, analytical characterization, air-sensitive techniques, and laboratory safety, sharing practical expertise and enabling independent research.

Indian Institute of Technology Kharagpur, India *(MS Researcher)* **2019 – 2020**

- Performed extensive reaction optimization and substrate-scope evaluation for organophotoredox-mediated amide and benzothiazole synthesis (*Chem. Eur. J.* **2020**, 26, 3703–3708).

Nanyang Technological University, Singapore *(Research Intern)* May **2019** – July **2019**

- Investigated an osmium high-nuclearity carbonyl complex as a potential photoacoustic imaging contrast agent and studied its reactivity with gold electrophiles (*J. Clust. Sci.* **2021**, 32, 929–935).

University of Victoria, BC, Canada *(Research Intern)* May **2018** – July **2018**

- Developed synthetic routes to carbazole-bis(tetrazole) ligands and evaluated the corresponding platinum and palladium complexes as sustainable oxidation catalysts (*J. Coord. Chem.* **2021**, *74*, 983–1008).

Indian Institute of Technology Kanpur, India *(Research Intern).* May **2017**– July **2017**

- Synthesized a hexatopic triptycene hexa-acid linker and investigated metal-assisted self-assembly into honeycomb-structured metal–organic frameworks (MOFs).

SELECTED PUBLICATIONS

1. **Samanta, S.**; Patel, L.; Manna, D.; Ojeda, M.; Powers, D. C. Base-Free C–N Cross-Electrophile Coupling, *J. Am. Chem. Soc.* **2026** *148*, 33104–33117.
2. **Samanta, S.**; Tan, H.; Chatterjee, S.; Powers, D. C. Synthesis of *N*-Pyridinium Aziridines. *Org. Synth.* **2026**, *103*, 78–100.
3. Roychowdhury, P.; **Samanta, S.**; Brown, L.; Waheed, S.; Powers, D. C. Bidirectional Electron Transfer Strategies for Anti-Markovnikov Olefin Aminofunctionalization via Arylamine Radicals. *ACS Catal.* **2024**, *14*, 13156–13162.
4. **Samanta, S.**; Biswas, P.; O'Bannon, B.; Powers, D. C. *β*-Phenethylamine Synthesis: *N*-Pyridinium Aziridines as Latent Dual Electrophiles. *Angew. Chem. Int. Ed.*, **2024**, *63*, e202406335 (Hot paper). *Top viewed article in* **2025**.
5. Tan, H.; **Samanta, S.**; Maity, A.; Roychowdhury, P. Powers, D. C. *N*-Aminopyridinium Reagents as Traceless Activating Groups in the Synthesis of *N*-Aryl Aziridines. *Nat. Commun.*, **2022**, *13*, 3341. Highlighted in *Synform* **2022**/11, A169–A171 as literature coverage.

Complete publication list available on Google Scholar.

SELECTED POSTER PRESENTATIONS

1. *"Base-Free, Cross-Electrophile C–N Coupling with Exclusive Cross-Selectivity."* at Gordon Research Conference on Organic Reactions and Processes — July 2025
2. "*β-Phenethylamine Synthesis: N-Pyridinium Aziridines as Latent Dual Electrophiles.*"
   a) Welch Conference on 'Frontiers in Molecular Catalysis' — Oct 2024
   b) Graduate Research Symposium by ACS Division of Organic Chemistry at Virginia — July 2024
3. "*Development of 1,2-Aminofunctionalized Library from N-Pyridinium Aziridines.*"
   a) The TexSyn Conference at TAMU — June 2024
   b) F.A. Cotton Medal for Excellence in Chemical Research Symposium at TAMU — April 2024
   c) FloHet-2024 at University of Florida — March 2024

ORAL PRESENTATIONS

1. "*N-Pyridinium Aziridines: Synthesis and Reactivity*." at Pfizer Chemistry Connect Virtual Symposium — Nov 2024
2. "β-*Phenethylamine Synthesis: N-Pyridinium Aziridines as Latent Dual Electrophiles.*" at Southwest Regional Meeting of the American Chemical Society — Oct 2024

HONORS & AWARDS

- Selected participant, 74th Lindau Nobel Laureate Meeting in Chemistry (2025)
- Derek and Christiane Barton Graduate Endowment (2025)
- Sharon Dabney Memorial Scholarship (2025)
- F. A. Cotton Endowed Memorial Graduate Travel Award (2025)
- Graduate Student Research and Presentation (RAP) Travel Award (2025)
- 3rd place in the poster presentation, Welch Conference (2024)
- Grand Prize for the best poster in Joint Research Symposium, TAMU (2023)
- Martin Donald Corera Memorial Travel Award (2023)

ORGANIZATIONAL LEADERSHIP AND VOLUNTEERING

Phi Lambda Upsilon (Honorary Chemical Society), TAMU *(Active member and ex-treasurer)* — 2022 – present

- Served as treasurer and chair of the judiciary board, managing financial operations and supporting the society's activities.
- Volunteered for Chemistry Open House at Texas A&M University (2022 and 2023) to promote chemistry education and engage the broader Aggie community.

ACS Division of Chemical Health and Safety Workshop *(Moderator)* — Aug 2023

- Introduced the RAMP (recognize hazards, assess risk, minimize risk, and prepare for emergencies) paradigm through case studies and interactive discussions to develop effective laboratory safety practices.

Graduate Student Association of Chemistry (GSAC), TAMU ***(At** large representative)* — 2022 – 2023

- Volunteered at various social and professional events to support, engage, and advocate for graduate students within the Chemistry department.
- Mentored prospective graduate students during Graduate Recruiting Week (2022–2023), facilitating their transition and providing insights into the graduate experience.

Technology Transfer Group, IIT Kharagpur *(Coordinator)* — 2018 – 2019

- Organized multiple events to promote intellectual property awareness and patent drive in IIT Kharagpur as the bridge between the research labs and industry.